

Active Inference BookStream 002.02 ~ Parr, Pezzulo, Friston ~ Chapters 4,

BookStream | Parr, Pezzulo, Friston ~ Chapters 4, 5, 7, 8 | 1:31:11

Hello, it's July 28th, 2023, and we're in Active Inference Textbook Group slash Bookstream 2.02.

Thanks, Ali, for joining.

So what we're going to do today is give a short overview of the chapters from the Parr et al. 2022 book.

We're going to do chapters 4, 5, 7, and 8.

And we're just going to pause between them because then we'll clip them into the shorter videos, append that to the playlist,

just so there's a first video overview of each of the chapters, and this is the second in that work.

All right, so we'll do chapter 4.

We'll just wait a few seconds and then start chapter 4.

Okay, chapter 4 is called The Generative Models of Active Inference, and it begins with a quotation, everything should be made as simple as possible but not simpler by Albert Einstein.

Ali, what is your overview, thought, or warning for chapter 4?

Okay, so after the preliminary materials in chapters 2 and 3, which was basically largely based on providing some conceptual framework for developing the further theory,

chapter 4 delves into much more detail in terms of mathematical formulation,

and it unpacks a lot more the way that the central equations of active inference is derived,

and how to construct the important elements of active inference models.

So, say, matrices A, B, C, and D, and also how to put together generative models in different situations.

So, it basically lays out the foundation for constructing active inference models,

both for discrete time situations and continuous time ones,

which will be used later in chapters 7 and 8.

But this is probably one of the most challenging and at least mathematically dense chapters in the book.

So, I would personally suggest reading through this chapter really slowly

and even if we don't get to understand every single detail of the chapter,

obviously we can return per required as we go through the textbook.

Thank you, Ali.

Yes.

So, let's look through the sections.

Just to add on, though, chapter 4 is one of the larger and more equation-dense chapters

because it is the common kernel or basis that's then going to get applied in chapter 5 in the neurobiological case.

There's a recipe for making chapter 4 in chapter 6.

That's the recipe for active inference modeling.

Chapter 7 and 8 are about the discrete and the continuous time variant or subtype or motif of these kinds of things called generative models.

So, this is the real common root and we'll just look at what the sections are.

This chapter complements the preceding chapter's conceptual treatment of active inference with a more formal treatment.

Section 4.2, from Bayesian inference to free energy.

What would you say about this section, Ali?

Okay, so, as we know, the Feyenoord principle is inspired by previous work on Bayesian inference.

I mean, all the way back to Helmholtz's theory about unconscious inference or something to that effect.

I can't remember the exact term.

I can't remember the exact term.

But here, I mean, it provides in a bit more detail how we can derive free energy principle formalism using the established Bayesian inference formulation.

And particularly, one of the key movements, or at least one of the key decisions through the derivation of free energy principle formulation

is using Jane's inequality principle to derive an upper bound instead of just using the exact values to compute or to achieve required parameters.

So, that's basically, in my opinion, the key premise of Section 4.2.

And to see in a bit more detail how we can achieve those upper bounds using Jane's inequality directly from, by using, I mean, manipulations of Bayesian inference, Bayesian statistical formalism.

Thanks.

I'll just add one point from this section.

Broadly, these are the problems of inferring states of the world, perception, and inferring a course of action planning.

So, this is, again, referring to the perception and action, and everything that happens in between is the internal or the cognitive part of the inference.

But this is like the blanket state, cybernetic input-output.

And then, let's look at the first equation, or how much equations overall, or what equations do you think we should highlight?

Okay, so maybe we can, I mean, just as a general comment about these different equations, well, each of these equations provide a distinct step toward deriving the ultimate whole picture.

So, even if we don't quite understand how we can derive from each step to the other one, it's good to know that it's only required to understand how we get to that ultimate whole picture.

But ultimately, what we would need in order to develop active inference models is the ultimate equation or ultimate whole picture.

So, this is just a way to elucidate the steps toward developing that whole picture.

But, again, it's not an essential requirement to understand the materials of the rest of the book.

But if we go from, I mean, equations 4.1 toward the 4.4, or other words, the variation of free energy, well, equation 4.1 is just a basic definition of some properties of probabilities in terms of conditional probability and so on.

So, equation 4.2 provides the central Jain's inequality principle and how it relates to, I mean, conditional

probabilities and, of course, joint probabilities.

And then, by using those two properties or those two equations, we ultimately get to 4.4, which is the definition of variation free energy parameter,

which is the parameter of interest that needs to be optimized in order to inference to happen,

or at least perceptual inference to happen in active inference models.

Thanks.

Only thing I'll add is F is the letter used for variational free energy.

You can think of it like a computer program, and the arguments that it takes in or the variables that it takes in are Q ,

which is the distribution that's under the statistician's control,

and Y , which are the data which are outside of the statistician's control.

And do you want to describe more about anything in this equation or carry on?

Just one thing that can probably be helpful is to somehow compare these steps with the initial picture we had from Chapter 2,

because variation free energy was first introduced in Chapter 2.

So it can be helpful to go back and forth between Chapters 2 and 4 and try to connect the dots between the related points there.

Section 4.3 Generative Models

All right, I'll read the first sentence, then you can give some thoughts.

To calculate the free energy, we need three things.

Data, a family of variational distributions, and a generative model comprising a prior and a likelihood.

Data, a

so clearly we would need two different types of generative models for each situation

and obviously the generative models or the way to construct generative models for

discrete time situations would vary quite a bit from the one for continuous time situations

but the general principle underlying those generative models are basically the same

which is i mean to somehow construct a model of the environment i mean either be it

for the situation that that is sequential in time or for the situations that need to be

somehow each moment of the situation needs to be accommodated in terms of a continuous time

situation so figure 4.2 provides some examples of both so uh for uh yes let me see yes uh so

uh we have some examples of different kinds of uh generative models some uh case case studies uh if

you like uh and it provides various ways to um show how the the dependencies between between variables can

be modeled using um these kinds of graphical probabilistic models so uh one common way to

represent generative models is to use these kinds of uh graphical probabilistic models in active inference literature uh which is uh which is uh at least in this case uh the circles would represent uh the random variables

uh and uh the squares that would represent the distributions uh which would describe uh the dependencies between

between those random variables uh so uh between those random variables uh so we can see uh the clear relationships between uh those parameters here which is basically what uh this whole graph uh what uh constitutes uh the generative model that needs to be used for um different situations and then

uh in figure 4.3 we can compare uh the two different types of generative models based on uh whether it's discrete time or continuous time situations so the upper picture is a generative model for the discrete time situation and the lower picture uh is uh the parallel continuous time version of it and as we can see uh the general topology of these um models are the same uh the only things uh that differ uh is the use of parameters uh for policies or i mean discrete time policies or the continuous time ones and we can obviously compare uh the different elements uh for beta for both uh priors states and uh external states internal states and so on by comparing uh these two models here yeah we

often return to figure 4.3 it's kind of the rosetta stone of generative modeling for the context of this book because it's then going to develop out into chapter 7 and 8 and it represents a really fundamental decision made in modeling and in the later chapters it's also shown how it can be made into a hierarchical model that combines aspects of both but within each level of modeling still these are the kinds of decisions that modelers are presented with when it comes to statistical modeling overall so section 4.4 goes into essentially the top half of figure 4.3 discrete time what would you say about discrete time okay so uh the discrete time situation uh is uh obviously the archetype uh discrete time situation which is the palm uh palm dp uh models so at this point uh i would uh very much like to recommend uh following uh the steps uh following the material from uh step-by-step uh paper because uh that's

in that paper uh the way to construct pump dp models uh is described in a bit more detail so uh if anyone feels

like they should uh like they should uh learn a bit more about uh the uh the gaps in the details uh i i would very much like to recommend that um particular paper uh so uh yeah uh i don't know how much detail uh we should

go into because uh it's uh i mean uh although uh it's not maybe uh detailed enough for some tastes but it goes in a quite uh extensive detail uh about how we can uh construct uh these models using uh the concepts we've learned uh in previous chapters uh so um

uh so um ultimately we reach equations 4.13 and 4.14 uh which are basically the culmination of palm dp formulation using uh the vector notations and gradients and so on uh so

So then we go to continuous time situation.

Great.

A few things intervene in the continuous time chapter.

Ones that we'll just mention here because they're kind of boxed or partitioned from the continuous time part,

but they're following pages.

First is Markov blankets.

We won't go into it here, but kind of footnote that or look at some other places where we talk about it outside of this chapter overview.

Figure 4.4, Bayesian message passing.

Again, a big topic.

Let's kind of just go past it now.

Back to the regularly scheduled continuous time generative model discussion.

And then another box to the generalized coordinates of motion.

So taking position plus derivatives of position.

And that has some beneficial properties that are described and unpacked also elsewhere.

Do you want to say anything about 4.5.2?

Well, the only thing that comes to mind is, although, as I said before, all the formulations here may look more, but I mean, a bit too dense to understand at the first pass.

But some of the key components here could be, obviously, the material from box 4.2 and 4.3,

I think are quite essential to understand the underlying principle behind deriving the continuous time situation.

Because without Laplace approximation, what we would have in terms of free energy minimization is, it would look very much like the Gibbs free energy.

So, I mean, the key distinction between the free energy principle, as described in active inference literature, as opposed to Gibbs free energy, is the distinction, is the Laplace approximation.

So, this is what enables us to go from Gibbs free energy to, I mean, the variation of free energy.

So, yeah, that's, I mean, quite essential to make this, to be familiar with this essential approximation.

And, obviously, the concept of generalized coordinates of motion will come time and time again throughout the whole book,

particularly in chapters 8 and 9.

So, yeah, those two concepts, I believe, needs a bit more attention.

And, um, so...

Yeah, sounds good.

Box 4.3 Laplace approximation.

Equations.

Another message passing.

Representation.

And a summary.

The key message to take away is that approximate Bayesian inference may be framed as minimizing a quantity

known as variational free energy.

This depends on a generative model that expresses our belief about how data are generated.

Anything else you want to add?

Uh, nothing comes to mind at the moment because, uh, as I said, this is, I mean, we're still, uh, in the stage that we want to develop our essential tools to be used, uh, in the rest of the book.

So, uh, here, up until now, I believe, uh, by the end of chapter 4, we have acquired all the essential necessary mathematical tools.

Uh, and the next chapter, chapter 5, uh, is, uh, kind of acts like an interlude.

Uh, and I don't think it's the direct, um, I mean, continuation of chapters, uh, one through four.

So, I believe the, the first, um, um, section, or the first part of the book, uh, conceptually and mathematically ends here.

So, yeah, that's it.

Yes, it's a little bit like the pragmatic modeling part gets foreshadowed or explored in five, now that we're all built up with four.

All right, that's the end of the overview for four.

Okay, chapter 5 is called Message Passing and Neurobiology.

What is your overview thought on chapter 5?

Okay, uh, I mean, uh, it's a kind of, I don't know, I have mixed feelings about this chapter because, uh, on one hand, uh, you see, as far as I understand, um, active inference, uh, although it originated, uh, as, uh, quote-unquote a unified theory of the brain, uh, I don't think, uh, it's a neurobiological theory per se.

Okay, uh, of course, uh, of course, there can, uh, be some correlations between neurobiological, uh, components or concepts, uh, with, uh, active inference, uh, active inference, um, I mean, concepts.

But, uh, I mean, uh, it's not, uh, an essential premise of active inference theory to provide, uh, a theory, uh, uh, to provide a comprehensive theory about how the neurobiology, uh, of human brain or, uh, other organisms' brain, uh, behave, uh, at a detailed and neuroanatomical, um, level.

Uh, but, uh, uh, then again, uh, it's nice to have these kinds of empirical correlations, uh, between the findings of neurobiology and the active inference, uh, theory, uh, but, uh, I, I don't think it's, uh, one of active inference, uh, central assertions, at least, uh, to my understanding.

Well said.

Very interesting framing.

Well, chapter five definitely takes a very specific system of interest approach by highlighting one of the most studied areas, also one of the most relevant areas, which is mammalian neuroscience.

And the chapter is going to introduce a few different motifs in the nervous system and essentially build up towards figure 5.5, which is at the end of the chapter.

And 5.5 wires together three specific neural systems that the chapter is going to focus on work in that area from.

So, Ali said it very well.

Active inference was built up to in chapter four.

Here is another level or type of science with assertions or with representations or mappings to any specific system.

But this is the kind of modeling that has been built up and done by Friston, Parr, Pizzullo, and others over the decades with a focus coming from a human neuroimaging laboratory setting.

A lot of focus and study and attention and funding and everything on the mammalian nervous system.

But claims about the nervous system have have are not the basis of what active inference claims or how it's derived.

But this is like an example case study in neurobiology connecting back to some of the formalisms that we've just seen introduced in chapter four.

Yeah, and to add a minor point to which you just said, I think it's important to draw attention to the last sentence of the last paragraph of the first page.

It is important to draw a distinction between a principle, i.e. the minimization of free energy and a process theory about how this principle may be implemented in a certain kind of system.

So I think this sentence here frames this chapter in relation to all the other technical chapters of this book.

So if every other chapter is about developing, or at least up to now was about developing the principled formalism of active inference, now chapter five provides a kind of preliminary sketch for the process theory of active inference, which is obviously far from an extensive theory.

It's just a single chapter, but then again, it can provide some important signposts for anyone who wants to further investigate this area.

Awesome. Free energy principle, Bayesian mechanics, all things in that area are on this principle, not responsive to empirical data.

And then the process theory is about how the principle is implemented.

So the specific generative models that are made and how well they map or how well they do in a portfolio of models that can have very different goals and assumptions and all of this.

But the process theory implementation lets us develop hypotheses that are answerable to empirical data.

Like, what is the kind of information or relationship between photons hitting the retina and changes in activity in neural systems?

And that's an informational question.

And that's an informational question or can be abstracted in a way to an informational question that it turns out does have empirical support and results in unique explanations and predictions.

That doesn't mean that it always results in unique explanations and predictions, but a lot of citations are provided here.

That's what we can explore in chapter five.

The last paragraph of the first section describes that they're going to look at the three different neural systems.

Okay.

Section 5.2, microcircuits and messages.

What do you think, Ali?

All right.

The ones we've seen before in chapters one through four.

I think it was in chapter four.

We can see some clear parallels between how this kind of cortical message passaging happens in the brain versus how it is framed in active inference literature.

And as we can see, it's clearly inspired by the neurobiology of the brain.

But then it's important to keep in mind that it's not a direct one-to-one mapping between these two models.

And so, this is just a kind of, I don't know, an interesting or illuminating, if you like, parallel to keep in mind to somehow be a bit more confident about the viability of the theory we want to use for message passing in active inference.

Which is to say, which is to say, which is to say, it's not some haphazard theory that's just been developed for practical reasons.

It has some basis in neurobiology, although it's not necessarily fully congruent with every detail of neurobiology.

Great.

Great.

This specific example is going to involve this one region of mammalian cortex tissue that has these six layers. And there's a ton of neurobiology.

So, the big takeaway for figure 5.1 is that it's possible to graphically lay out nodes and variables and find some empirical correspondences.

Again, some unique explanations and predictions in certain cases.

And that's one kind of modeling where it's really trying to understand and improve the ability to do correlation and intervention and counterfactual causal type analysis with the real system of interest.

Or, in a more pedagogical setting or in a more pedagogical setting or in a more pedagogical setting or in a more pedagogical setting or in a more pedagogical setting, you might sweep across large families of structures of models.

And there's no need to be grounded to any biological structure at all.

So, this is just describing the specific neuroanatomical research that really arose out of the imaging work at UCL and SPM package.

So, that's where a lot of this comes from.

5.2?

Yeah, go ahead.

And sorry, just as a side note, I think watching one of Thomas Parr's lectures on neurobiology of active inference, which is available on YouTube, would really help to understand the materials of this chapter better.

So, I highly recommend watching that one.

Thanks.

Figure 5.2.

Give a re-rendering of a kind of classical view of a hierarchical predictive coding system works.

So, here, abstracting a layer from the tissue 6 layer to just two layers here.

Computational layers now.

And then showing how there's hierarchical communication within a layer, but also there's signaling within a layer.

And there's a hierarchy in Bayesian modeling with variables that are higher order predictions about other variables.

And that's the basis of the predictive coding architecture.

So, 5.2 looks at some ways that the, something that resonates with the cerebral cortical architecture enables what might computationally look like or have some really strong and explanatory values in actually relating to

computationally a hierarchical Bayesian model, which could do various general tasks.

All right, 5.3 is motor commands, leaving the prefrontal cortex going down to the butterfly-looking cross-section here.

what is 5.3

okay so

5.3 moves to

the other half of active inference

framework which is

I mean

how it can model

the decision making

and ultimately

the movement of the agent

in order to

minimize the

expected free energy as opposed to

variation of free energy

that we saw in perceptual

half of active inference

so

it again provides

a kind of

correlation or

analogy between the structural

neuroanatomy

particularly related to

I mean the motor

commands and

how it

can relate to active

inference and particularly

the continuous time active inference

so

we can see that

for

I mean for the external

event

or I'm sorry for the external

state we can take
for example the
proprioceptive afferent and then
this
proprioceptive afferent
acts as a kind of
why for
the continuous time active
inference which needs to be
I mean
processed in
a way to optimize the
expected free energy
and how it relates to
both attention
and precision
we will see a bit more
detail about
those
those terms and
the relation between them
in chapter
eight but
I think
here
section 5.3
provides a good summary
about the general
paths through
the motor
command systems of
neurobiology
great
I'd say
while
the
previous case study
focused on how the

connectivity
within and between
the cortical columns
could have a
computational
relationship
with a
Bayesian
hierarchical
predictive coding
architecture
the argument of
this second case study
is that a
continuous input
continuous output
kind of
set point
seeking
reflexive
motor behavior
with a
moving set point
with a
descending
moving set point
enabling motion
by changing
ultimately the set point
and enabling
variation
in the strategies
to reach that set point
through
different mechanisms
this
is also
describable

in a
compatible
way
that's a
shorter section
now section
5.4
subcortical
structures
what would you say
about this section
okay so
subcortical
structures
are
very important
in
the
the decision
making
and
I mean
the planning
of the
agents
so
obviously
here
we need
to
we need
another
kind of
analogy
between
the way
that
these

plannings
and
decision
makings
happen
neuroanatomically
with the way
that
it's framed
in active
inference
but again
we can see
it's
clearly
based on
at least
some of
the
important
elements
we've
seen
from
the
previous
chapters
so
for example
we saw
how
policy
is described
or how
it relates
to outcomes
and preference
and so on

we can see
those elements
are
directly
inspired
by
neuroanatomical
structures
so
I guess
that's
at least
in my
opinion
this section
here
5.4
seems a bit
more
sketchy
in the
meaning
that it
doesn't go
into
quite
the extensive
details
about
how
those
structures
can be
compared
but
for anyone
who wants
to

further
investigate
these
topics
there are
some
useful
references
put on
here
on pages
93
and 94
so
yeah
thanks
yeah
it's
really
abbreviated
and
overviewed
but
we get
an
interlude
from
table
5.1
with
putative
roles
of
neurotransmitters
so
same
perspective
that we

took
before
on
neuro
anatomical
functionalism
here
directly
translates
to
neurotransmitter
reductionism
or essentialism
or something
like that
so
certainly
all
neurotransmitters
and molecules
they play
variable
roles
in different
settings
and
this is
the neat
and scruffy
manifold
all over
again
one person
might say
well
we need
a theory
for every

acetylcholine
molecule
in the world
they're all
in a unique
context
and someone
else says
all neurotransmitters
are described
by one parameter
in this model
I'm getting
value from it
so to me
that's an
account
and
somewhere
in between
is
the work
in this
space
which is
making
an attempt
to have
a principled
and falsifiable
approach
to
model
the
computational
aspects
of specific
regions

and contexts
and settings
and so
acetylcholine
noradrenaline
dopamine
and serotonin
are
given a little
mini review
here
and so
it's not
an exhaustive
or an
exclusive
claim
it's kind
of a
provocation
from
computational
and molecular
neuroscience
and people
can look
into the
papers
and also
ones that
probably
have been
published
since
5.6
goes to
continuous
and discrete

hierarchies
which
is graphically
overviewed
in figure
5.5
so what
would you
say about
this
yeah
one
interesting
thing
about
this
section
is
the
observation
that
our
lower
level
engagement
with the
environment
can be
most
successfully
characterized
with continuous
time
formulations
but as
we go
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level
of
cognitive
concepts
or
at the
level
of
cognitive
hierarchies
and
we
come to
concepts
such as
I don't
know
decisions
or
even
beliefs
and so
on
we can
reach
the
area
that
the
discrete
time
situations
would
probably
be
more

efficient
to
characterize
the
behavior
of the
agent
so
this
multi-scale
structure
of
active
inference
modeling
is quite
evident
in
the
way
that
our
message
passing
happens
in our
brain
in terms
of
lower
level
data
processing
up
into
consolidating
the
higher

level
cognitive
concepts
and
ontologies
awesome
thank
you
to me
figure
5.5
demonstrates
the kind
of
whole
of
body
approach
that
you could
imagine
there's so
many organs
and
systems
and
phenomena
for which
there aren't
specific
generative
models
so little
can be
said about
situations
where no
generative

model
has been
articulated
and
here's
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has
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gives
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also
it's
kind
of
like
reading
a
Drosophila
melanogaster
review
paper
relatively
it's
like
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is
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much
work
it
takes
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state
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knowledge
in
an
insect
so
then
in
another
insect
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about
that
insect
empirically
and
genetically
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consider
this
to be
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sophisticated
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specific
cognitive
systems
at least
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reflexive
aspect
to this
chapter
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of
active
inference
but
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it's
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inference
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seven
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chapters
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chapter
eight
on
discrete

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forks
of
a
river
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discussed
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group
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animals
to
one
level
deeper
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classification
scheme
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way to
the
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eight
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of
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uncertainties
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least
in a
practical
sense
so
I
believe
these
two
chapters
are
really
helpful
in
order
to
consolidate

our
understandings
from
the
previous
chapters
awesome
well
said
so
it's
going
to
involve
specifying
some
discrete
time
models
7.2
goes
into
perceptual
processing
and the
general
structure
of the
chapter
is
going
to
walk
through
a
series
of
examples

that
build
in
complexity
where
they
first
start
with
perception
in
7.2
introduce
decision
making
and
then
describe
a few
more
types
of
motifs
or
cognitive
structure
or
patterns
and
also
check
out
step
by
step
and
model
stream

one
where
it's
built
up
to
in
a
different
way
so
the
first
example
is
I'll
let
you
describe
it
since
it's
musical
okay
so
yeah
the
first
example
is
the
situation
in
which
we
try
to
describe

the
performance
of an
amateur
musician
in terms
of
how
we
listen
to
the
performance
of an
amateur
musicians
in terms
of
the
predictions
we
get
from
our
anticipation
of
the
following
notes
as
opposed
to
the
actual
notes
that's
being
played

so
these
kinds
of
anticipatory
reaction
listening
reaction
to
the
musician
can
be
successfully
formalized
using
discrete
time
active
inference
by
putting
up
together
the
matrices
A
for
the
states
and
matrix
B
for
the
transition
between
the

states
or
the
transition
probabilities
which
in
this
case
describes
the
probability
from
going
from
one
note
to
the
other
and
obviously
the
actual
sequence
that's
been
played
which
can
be
described
with
the
matrix
D
so
and

another
point
I wanted
to
mention
is
for
anyone
who
has
downloaded
this
chapter
before
I don't
know
I think
about
June
or
something
I
recommend
re-downloading
it from
MIT's
website
because
they have
corrected
some of
the
typos
that
was
previously
present
in

this
chapter
particularly
in
figure
7.2
cool
so
this
graphical
model
where
a
person
is
listening
this
is
a
general
perceptual
Bayesian
framing
it's
specified
just like
with any
other
equations
there's a
lot to
look
into
but
A
indicates
the
probability

of an
outcome
given a
state
this
is
saying
if
it
were
all
on
the
diagonal
like
an
identity
matrix
this
is
kind
of
a
common
motif
then
states
kind
of
map
to
themselves
so
in the
context
of
in the
context

of
this
model
A
represents
the
mapping
between
the
observed
note
and
the
underlying
hidden
true
note
B
describes
the
transition
matrix
of how
those
change
to
time
D
is
the
prior
they're
specified
figure
7.2
do you want
to describe
it

all right
so in
figure 7.2
or at least
the incomplete
version of
figure 7.2
we see here
well
at the
upper
left
part of
the
picture
we see
I mean
the
beliefs
about
each
note
at
each
step
at
each
time
step
and
at
upper
right
we
somehow
translate
those
beliefs

into
specific
numerical
values
so
instead
of
just
assigning
some
continuous
values
values
we
we've
simplified
the situation
by assigning
some
discrete
numerical
values
for
each
note
and
then
the
lower
left
is
supposed
to
show
the
free
energy
gradients

over
time

or
in
other
terms
the
prediction
errors
we
get
from
I
mean
comparing
our
predictions
with the
actual
outcomes
so
lastly
the lower
right picture
shows
in parallel
to the
upper right
picture
determines
the
values
of
these
errors
so
we

can
see
both
the
continuous
the
initial
or at
least
initial
continuous
assignment
and values
and then
the
further
discretizing
of the
values
in order
to
get
the
discrete
time
situation
or
the
more
tractable
discrete
time
situations
okay
so
it's a
general
passive

inference
task
where
there's
priors
about
how
states
are
going
to
change
through
time
and
then
there's
real
data
coming
in
so
that's
the
kind
of
classical
predictive
coding
video
compression
Kalman
filter
Bayesian
setting
7.3
introduces
a key

motif
which is
decision
making
and
planning
as
inference
so
this
is
the
idea
of
having
a
Bayes
graph
where
the
variables
can
relate
to
different
things
there's
high
composability
and here
the idea
is that
a variable
is going
to be
proposed
that we
can do

inference
about
that
describes
the
process
of
decision
making
or
policy
selection
so
what
would
you
say
about
7.3
okay
so
7.3
is
obviously
similar
to
what
we
saw
in
chapter
4
and
if
I'm
not
mistaken
even

the
topology
is
exactly
the
same
with
that
picture
we
saw
previously
so
this
is
the
initial
setup
to
or
which
acts
also
as a
review
about
how
these
different
components
of
POMDP
generative
models
needs
to
be
described

in
such
situations
but
ultimately
the
specific
case
study
we
come
across
in this
section
is
the
attempt
to
model
the
behavior
of
a
mouse
in a
teammate
so
a
rat
and
a
teammate
so
especially
teammates
containing
an
aversive

stimulus
in one
arm
and
an
attractive
stimulus
on the
other
so
this
can
act
as
a
kind
of
toy
example
to
use
this
kind
of
probabilistic
modeling
to
describe
these
situations
thanks
so that
leads us
right to
figure
7.4
here's a
visualization

of the
situation
with
the
rat
in this
case
where
there's
a
pleasant
and
aversive
stimuli
on each
end
of a
decision
point
and
there's
also
a
epistemic
opportunity
to
receive
some
information
about
the
context
that
the
animal
is
in
and

so
that
setting
is
described
for
both
the
case
with
white
on the
left
black
on the
right
and
black
on the
left
white
on the
right
and
those
are
shown
in
terms
of
their
differences
in
the
matrices
the
explicit
specification

of
the
generative
model
visualizations
show
some
of the
slices
of
the
B
variable
which
reflect
different
transition
probabilities
C
represents
the
preferences
which
are
expressed
over
the
observable
states
D
reflects
the
priors
on the
different
states
that need
priors

7.4

what would

you say

about this

okay

so in

7.4

it builds

up on

the

previous

section

and

adds

other

elements

that we

previously

saw

in

chapters

3

and

sorry

2

and

4

which

is

how

the

exact

formulation

for

expected

free

energy

can be

used
sorry
variation
free
energy
can be
used
to
formulate
the
trade-off
between
the
information
seeking
or at
least
between
the epistemic
value and
information
seeking
so
here
it uses
again
that rad
example
in a
bit more
more
extended
and
elaborate
form
to
formulate
the

epistemic
value
of
observing
q
in a
given
location
and
figure
7.7
is a
representation
of
this
situation
but
another
situation
that's
been
let me
see
yeah
in
7.9
another
case
study
discussed
here
is
the
situation
of
the
saccotic
eye

movements
and
because
it is
something
that can
be
quite
successfully
described
or characterized
in terms
of
information
seeking
versus
the
epistemic
value
and the
situation
here
is
shown
visually
in
figure
7.9
which
clearly
shows
how
our
visual
saccadic
eye
movements
can be

described
in such
a way
as to
trace
the
trajectory
of our
eye
movement
among
different
regions
of the
visual
space
and how
the
information
we
gather
from
a given
region
can
affect
the
subsequent
trajectories
of our
saccadic
eye
movement
so
yeah
that's
basically
the

main
premise
of
this
section
I guess
nice
great
7.5
what
would you
say
about
it
okay
so
7.5
again
adds
another
dimension
to
the
previous
formulations
and
this
time
we
get
to
update
the
generative
models
by
learning
and

the
so
the
generative
models
for this
situation
is a
bit more
complicated
than the
previous
ones
because
it
now
needs
to
account
for
a
mechanism
or
a
way
to
update
the
matrices
we
had
before
so
in
the
previous
situations
we

didn't
account
for
learning
per se
but here
we
directly
update
our
sorry
the word
update
can be
confusing
here
we
get to
somehow
improve
our
generative
models
to
accommodate
for
these
updating
accounts
and
yeah
so
this
the
situation
here
or the
case

study
here
which
somehow
elucidate
the way
that
the learning
can be
accounted
for
with
these
models
is
again
a toy
example
of
a
creature
in a
simple
world
of
black
and white
tiles
which
kind of
tries
to
find
a
path
to
reach
a given

destination
a certain
destination
so
it is
more
complicated
than
the
situation
we had
for
the
rat
example
because
it only
had
I mean
simple
trajectories
that needed
to
traverse
but here
the
creature
or the
agent
in this
case
needs
to
do
lots
more
learning
and

information
seeking
and so
on
so
all
the
previous
elements
is
kind
of
combined
in
this
example
and
it's
a really
good
example
to see
how
the
different
components
of
active
inference
can
be
connected
to
each
other
nice
and
7.6

hierarchical

or deep

inference

first

a box

7.3

interlude

on

structure

learning

boxed

off

topic

and a

lot

to

say

but

structure

learning

broadly

refers

to

learning

the

structure

about

a

model

using

the

same

types

of

methods

that

you

might

to
do
inference
on
for
example
a
more
observable
sensor
data
reading
something
like
that
this
section
works
towards
the
idea
of
nested
inference
or
multi-scale
modeling
what would
you say
about
figure
7.12
okay
so
again
this
situation
is

I think
the most
complex
situations
of this
chapter
which
builds
up
from
the
previous
sections
and
this
time
it adds
another
layer
to
accommodate
for
the
inferences
that
happen
in
different
time
steps
so
in this
case
we have
multi-time
or
multi-scale
inference

and learning
happening
both
at the
levels
of
learning
and
at the
levels
of
information
seeking
so
this
this is
represented
in
figure
7.12
which
represents
how
kind of
this
fractal
generative
model
can be
seen
as
a
component
in
this
multi-scale
bigger
generative

or
as a
kind of
leaf
in
this
bigger
generative
model
so
it can
be
seen
as
a
lower
level
inference
happening
at the
leaf
level
going
up
to the
hierarchy
and
influencing
sorry
collaborating
and the
whole
process
of
learning
and
inference
at

the
higher
level
so
yeah
I guess
that's
somehow
summarizes
this
figure
so
if you
have
anything
to add
that's
that's
great
it's an
example
of the
composability
of generative
models
what
we've
talked about
and had
Toby
Sinclair
Smith
describe as
the
compositional
cognitive
cartography
and just

what
kinds of
connectors
can and
can't
you do
and how
can that
motif
that the
discrete
time
model
introduces
and then
the rest
of these
features
including
action
and learning
and so
on
get layered
in on
top
what can
you do
with that
713
gives
another
example
do you
want to
say anything
about it
or maybe

continue
on
yeah
so the
case study
here is
the example
of
linguistic
I mean
language
learning
through
reading
so
not language
learning
maybe
just
what happens
sentence
comprehension
yeah
yeah
in comprehension
so what
happens
when
reading
in an
anticipatory
way
the words
that
comes
each after
the other
so

why
this kind
of situation
can be
most
successfully
characterized
with this
kind of
modeling
because
it
involves
different
scales
of
learning
and
comprehension
both
at the
level
of
I mean
reading
at the
level
of
somehow
observing
the letters
and then
going
onto
the words
and then
word groups
and so

on
so
yeah
that's
a really
interesting
way to
again
combine
all of
those
elements
into a
single
unified
model
to see
how
those
different
time
scales
slow
and
fast
time
scales
operate
together
to
build
this
more
encompassing
model
of
more
encompassing

generative
model
of the
situation
great
any
closing
thoughts
on
seven
nothing
particular
not bad
thanks
all right
next
chapter is
chapter eight
which is going
to go into
the
continuous
time
all right
chapter eight
is called
active
inference
and
continuous
time
begins
with that
timeless
quote
everything
flows
nothing

stands
still
so
what would
you say
about
chapter
eight
all right
so
this
chapter
probably
is
my most
favorite
chapter
in the
book
because
of my
own
personal
interest
in
process
materials
and so
on
but
yeah
so
chapter
seven
acts
as a
really
good

starting
point
for
anyone
who
wants
to
develop
the
discrete
time
situations
to
model
discrete
time
situations
within
active
inference
framework
but
in
chapter
eight
we
kind
of
get
to
model
a bit
more
interesting
or
let's
say
more

involving
situations
and
they're
not
necessarily
kind
of toy
examples
we saw
at least
at the
beginning
of chapter
seven
so
obviously
as the title
suggests
this chapter
deals with
the continuous
time
situations
so
in that
case
we need
to
maybe
at this
point
refresh
our memory
about
what
continuous
time

situation
involves
by
reading
the
relevant
parts
reading
or reviewing
relevant
parts
of chapter
four
so
yeah
in chapter
four
we saw
that
the
generative
model
for continuous
time
situation
derives
from
the
ito's
stochastic
calculus
in terms
of
putting
the whole
process
into
two elements

two
stochastic
two elements
of stochastic
equations
one of which
is
the actual
states
the condition
of actual
states
or the
behavior
of the
actual
states
and the
other
one
is the
randomness
that we
need to
account for
in each
real-time
continuous
time
situations
so that's
what we
get here
in equation
8.1
and then
building up
from that

equation
we
it generalizes
that equation
to involve
I mean
the
functionals
of G
and F
instead
of just
the
single
valued
functions
of G
and F
so
then
we
get to
put
that
into
the
situation
that can
be used
for
describing
the
behavior
of
dynamical
systems
which is
a very

well-known
situation
to use
these
kinds
of
stochastic
equations
and
it's
widely
studied
how
those
kinds
of
dynamics
can be
characterized
especially
in
recent
Bayesian
mechanics
paper
by
Dalton
Saktivadev
and others
so
and then
it gets
to some
more
specific
examples
such as
Lotko-Volterra

dynamics
and
synchronicity
and so
on
in order
to show
how
these
kinds
of
dynamics
can be
elaborated
upon
and
can be
generalized
to
and
enables
them to
characterize
more complex
situations
so
yeah
that's
a really
short
and brief
overview
of the
whole
chapter
maybe
we can
talk about

a bit
more details
as we
go through
it
great
well said
well
I'm sure
for another
day
the
philosophical
implications
of
eight
seven
and
eight
and
high
road
and
low
road
and
all
these
other
parts
of the
textbook
great
topics
I
agree
I would
see

chapter
eight
as
demonstrating
continuity
with
some
classical
continuous
time
modeling
motifs
from
a few
different
areas
of
dynamical
system
science
which
is
applied
in
many
fields
but
these
are
some
classic
examples
so
figure
8.1
goes a little
bit more
into depth

or at least
into more
formalism
detail
about
exactly
what we
saw
in
chapter
five
with
the
spinal
reflex
arc
with
the
proprioceptive
data
coming
in
and
then
a
differential
being
calculated
with
the
set
point
which
reflects
a
descending
prediction
from a

decision
making
layer
and
that
can
be
viewed
as
this
kind
of
mechanics
that
plays
out
in
a
phase
space
in
continuous
time
like
a
spring
moving
around
with
someone
making
a
certain
path
within
a
tractor
and

a
spring
being
dragged
around
something
in
that
area
box
8.1
goes into
a very
fascinating
topic
do you
want to
describe
it
well
it's
maybe
one of
the
most
thought
provoking
pages
of the
whole
book
and
if I
remember
correctly
in
all of
the

cohorts
this
particular
box
I
mean
gives
always
gives
rise
to
lots
of
questions
because
of
some
of
the
interesting
and
at
least
initially
counter
intuitive
claims
here
but
I
don't
want to
spoil it
so
as a
spoiler
alert
it

kind of
gets
to
really
interesting
but
alas
very
brief
discussion
about
comparing
these terms
precision
attention
and sensory
attenuation
and
the
relation
and
similarities
and
difference
between
these
three
terms
and how
understanding
each of
them
is essential
to
understanding
the other
ones
but

as I
said
it's
a really
interesting
topic
which
gives
rise
to
lots
of
discussions
and
I
believe
it's
one of
those
topics
that
that's
worth
looking
a bit
more
looking
into
in
some
other
literature
as well
great
well said
what a
cliffhanger
next

they go
to
a
classic
model
family
called
Laca
Volterra
these
dynamics
inherit
from
characterizations
of
predator
prey
dynamics
in
ecology
so
it's
kind
of
a
classical
ecology
model
shown
in
figure
8.2
on
the
top
it's
actually
the

ecosystem
model
plants
herbivores
and
carnivores
which
follow
different
kinds
of
oscillatory
trends
in
continuous
time
and
so
that
also
has
enabled
it
to
be
applied
for
other
so
called
winnerless
competitions
and that
relates to
topics like
neural
darwinism
and also

neural
dynamics
where
things
have
kind
of
oscillatory
relationships
with each
other
which are
being
modeled
as a
continuous
time
underlying
process
with a
lot of
measurement
noise
and
discretization
through
space
and time
those are
the kinds
of
algorithms
that
SPM
explores
more
and
there's

lack
of
volterra
and a
lot
of
other
dynamical
systems
theory
in
SPM
so
active
inference
kind of
adds
action
and more
to
what was
laid out
from a
pure
dynamical
systems
theory
in
SPM
here
it
really
is
just
showing
the
ecology
example

and how
you can
project
if you
have
three
different
species
you can
think
about
that
motion
in a
cube
or
tetrahedron
and then
you could
project
onto
kind
of like
looking
at a
lower
dimensional
manifold
relating
just
two
of the
three
species
and that
evinces
this kind
of

oscillatory
but also
moving
behavior
that gets
connected
in
figure
8.3
to
neurobiology
what would
you say
about this
okay so
here in
figure
8.3
we see
some
applications
of
a lot
of
altera
dynamics
so
the
left
column
here
represents
what
happens
in
eye
blinking
eye

blink
conditioning
so
of
course
here
we
need
to
account
for
I
mean
the
expected
states
of
the
sequences
of
events
that
happens
in
the
eye
blinking
so
the
upper
left
figure
shows
the
expectations
in
terms
of

time
and
then
the
parallel
right-hand
side
equation
sorry
right-hand
side
figures
shows
the
lot
covalterra
system
that
is applied
in the
handwriting
situation
so
as we
can
see
although
the
I mean
mathematical
technology
is the
same
or at
least
the
modeling
technology

is the
same
the
outcome
of each
situation
varies
drastically
in
two
distinct

two
distinct
neurobiological
behavior
not
neurobiological
but
biological
behavior
so
yeah
we can
see
how
the
same
modeling
framework
can
give
rise
to
different
outcomes
based
on

what
parameters
needs
to be
optimized
what
parameters
are
selected
for
modeling
and so
on
so
I believe
it's a
quite
interesting
example
to compare
handwriting
and the
blinking
together
and
how
those
can be
compared
to each
other
using
the
latke
voltero
dynamics
great
thank you

box

8.2

gives a

variant

on the

learning

here

presented

with the

formalism

for

continuous

models

kind of

a

technical

aside

section

8.4

is about

generalized

synchrony

so figure

8.4

is going

to visualize

one of

the

classic

dynamical

systems

which is

the

Lorenz

attractor

so

what

would

you
say
about
this
figure
okay
so
this
section
is
truly
interesting
because
when one
thinks of
active
inference
probably
the first
situation
that comes
to mind
is
the
situations
in which
we have
quite
well
defined
probability
distributions
for
different
parameters
but as
we can
see

here
in
section
8.4
actually
some of
the
formalism
of
active
inference
can be
successfully
used
to
characterize
even
chaotic
systems
and in
particular
the way
in which
two
chaotic
systems
can be
synchronized
with each
other
so
this is
a classic
example
of
chaotic
Lorenz
system

and
it
derives
from
it draws
upon
from
some
of
Professor
Pristin's
earlier
work
on
birdsong
synchrony
and
as a
side
note
any
literature
before
2016
is
considered
earlier
history
in active
inference
literature
because
it evolves
quite
rapidly
so
this
kind of

synchrony
between
two
chaotic
systems
can be
interpreted
as
providing
evidence
or even
let's
say
a way
to model
a kind
of
primitive
theory
of
mind
in
the
sense
that
how
exactly
can
we
understand
or
can
two
agents
can
trace
each
other's

trajectories
without
engaging
in any
direct
exchange
of
observations
between
their
internal
and external
states
so
that's
a really
good
example
and I
believe
one of
the most
interesting
examples
of how
active
inference
can even
account for
these
kinds
of
behavior
so
and the
rest of
the section
goes into

the details
of how
this kind
of synchrony
between
multi
scale
Lorentz
systems
can happen
and how
can we
formulate it
mathematically
in terms of
continuous
time active
inference
awesome
and there's
been more
recent work
on
Markov
blankets
and
stochastic
chaos
but the
bird
example
is a
classic
8.5
goes into
hybrid
discrete
and

continuous
models
so
this
could be
kind of
like an
in-between
chapter
of
7 and
8
but now
that we've
been
introduced
to the
pure
form
of
discrete
and the
pure
form
of
continuous
models
here
is shown
that
that
composability
extends
to
so-called
hybrid
models
where

here
the
lower
level
visually
is using
the
continuous
time
formalism
and
the
higher
level
is
describing
the
discrete
time
formalism
and
this
was
the
similar
structure
described
by the
authors
of
paper
active
inference
does
not
contradict
folk
psychology

where
they
described
this
lower
level
as
motor
active
inference
which
was
closely
allied
with
the
spinal
arc
reflex
shown
above
and
then
this
higher
level
they
called
decision
active
inference
because
in
that
case
it
was
referring

to
a
discrete
decision
and
so
they
used
that
kind
of
basic
motif
of
continuous
activity
or
continuous
time
modeling
at
the
more
peripheral
aspects
of a
cognitive
entity
and
like
Ali
said
more
discretization
and
hybridization
as well
at

higher
levels
of the
cognitive
modeling
and
that
type
of an
architecture
here
instead
of
describing
who
wants
the
ice cream
cone
I
believe
here
it's
going
to
be
a
mixed
or
hybrid
model
that
is
going
to
call
back
the

ice
arcade
system
where
there's
a
fixed
point
that
is
able
to
be
moved
as
a
set
point
and
then
there's
a
continuous
time
ice
arcade
that
pursues
the
new
fixed
point
and
so
that's
analogous
to
a

new
set
point
or
fixed
point
being
specified
from
the
top
down
muscle
command
about
a
new
location
for
a
muscle

followed
by
movement
towards
it
this
is
a
muscular
activity
that is
realizing
that
but
but not
in the

elbow
coming away
from the
hot stove
this is
about
the
eye
circading
to an
epistemic
foraging
location
specified
by top
down
hierarchical
systems
8.3
describes
little
technical
aside
on
mixture
of
gaussian
gaussian
mixture
models
kind
of a
technical
modeling
note
and
8.6
closes

it
says
it's a
huge
topic
and
much
has
been
left
out
and
so
they
list
in
table
8.1
key
advances
in
continuous
time
models
and
those
areas
are
synthetic
birdsong
ocular
motor
delays
conditioned
reflexes
smooth
pursuit
eye

movements
psychosis
illusions
saccades
action
observation
attention
hybrid
models
and
self
organization
and
that's
chapter
8.
What
else
would
you
say
and
also
what
would
you
kind
of
lead
someone
to
in
the
philosophical
implications
of
8
because

it
sounds
kind
of
cool
okay
so
well
the
case
of
continuous
time
active
inference
inference
I
think
it
leads
to
really
interesting
questions
both
in terms
of
philosophical
questions
and also
more
practical
modeling
questions
about
what
parameters
needs

to be
accounted
for
and so
on
and as
I said
I
believe
it's
a more
more
interesting
way
of
if not
interesting
but at
least
more
involved
way
of
doing
active
inference
modeling
one
thing
that
one
of
the
philosophical
questions
that
Mao and
I

have
explored
in our
paper
is
how
the
processes
of
I mean
ontological
processes
can
philosophically
described
using
FEP
assertions
in terms
of
their
intra
action
with the
environment
in which
they
co-constitute
themselves
and
we don't
necessarily
distinguish
between
the internal
and the
external
states

so
one
obvious
example
of this
is
that
generalized
synchrony
example
that we
saw in
this
chapter
in which
we don't
necessarily
distinguish
between
which of
the birds
act as
the
agent
and which
one is
the
environment
or the
vice versa
so
these
kind of
co-constitution
of the
environment
and the
agent

which
gives
rise
to
the
partitioning
of state
space
through
Markov
blanket
is
one of
the
interesting
philosophical
points
that I
think
needs
to be
elaborated
a bit
more
using
some of
the
recent
advances
in
philosophy
such as
the tools
that's been
developed
in new
materialism
school

or some
other
philosophical
approaches
but
these
kinds
of
what
exactly
gives
rise
to
emergence
what
is
the
ontological
status
of
emergent
properties
and so
on
are
some
of
the
burning
questions
for many
philosophers
today
and I
believe
active
inference
and

particularly
continuous
time
active
inference
provides
a
clear
precise
mathematical
formalism
even
if not
to
answer
these
questions
but at
least
to
explore
it
in a
more
rigorous
and
practical
way
and
also
practical
and
attractable
way
so
this
is
the

area
that I
believe
philosophy
and
science
are
beautifully
intertwined
into a
coherent
view
of
not
only
the
phenomenon
of
interest
but
even
about
the
whole
world
wow
wow
pretty
cool
yeah
a lot
to say
about
that
topic
after
completing
chapter

seven
and
eight
you've
seen
the
kind
of
two
major
branches
or
two
major
motifs
of
just
one
kind
of
modeling
but
these
kind
of
models
have
so
many
different
forms
that
that's
why
it's
such
a
hands-on

process
to
specify
the
generative
model
in
chapter
six
and
fit it
with
data
in
chapter
nine
those
are
all
what's
required
and
that's
kind
of
the
last
mile
of
where
these
discussions
about
general
motifs
gets
you
but

also
playing
with
these
pedagogical
models
can be
really
helpful
basic
patterns
and
relationships
and
start
to
see
different
patterns
in the
graphical
models
and
know
from there
what levels
of
technical
processes
can be
kind of
coarse-grained
over
all right
okay
well
that's
it

I guess
next time
we will do
probably
nine
ten
and
maybe
something
else
all right
I'll end it
now
thanks Ali
thank you
thank you

thank you
thank you

thank you
thank you
thank you